# MATHEMATICS AND STATISTICS

**Time: 3 Hrs.** **Max. Marks: 80**

***General instructions:***

*The question paper is divided into **FOUR** sections.*

*(1)* ***Section A:*** *Q. 1 contains **Eight** multiple choice type of questions, each carrying **Two** marks. Q. 2 contains **Four** very short answer type questions, each carrying **One** mark.*

*(2)* ***Section B:*** *Q. 3 to Q. 14 contain **Twelve** short answer type questions, each carrying **Two** marks. (Attempt any **Eight**)*

*(3)* ***Section C:*** *Q. 15 to Q. 26 contain **Twelve** short answer type questions, each carrying **Three** marks. (Attempt any **Eight**)*

*(4)* ***Section D:*** *Q. 27 to Q. 34 contain **Eight** long answer type questions, each carrying **Four** marks. (Attempt any **Five**)*

*(5) Use of log table is allowed. Use of calculator is not allowed.*

*(6) Figures to the right indicate full marks.*

*(7) Use of graph paper is not necessary. Only rough sketch of graph is expected.*

*(8) For each multiple choice type of question, only the first attempt will be considered for evaluation.*

*(9) Start answer to each section on a new page.*

## SECTION – A

**Q.1. Select and write the correct answer for the following multiple choice type of questions:** **[16]**

i. The dual of statement $t \vee (p \vee q)$ is _____.

(A) $c \wedge (p \vee q)$ (B) $c \wedge (p \wedge q)$ (C) $t \wedge (p \wedge q)$ (D) $t \wedge (p \vee q)$ (2)

ii. The principal solutions of the equation $\cos\theta = \frac{1}{2}$ are _____.

(A) $\frac{\pi}{6}, \frac{5\pi}{6}$ (B) $\frac{\pi}{3}, \frac{5\pi}{3}$ (C) $\frac{\pi}{6}, \frac{7\pi}{6}$ (D) $\frac{\pi}{3}, \frac{2\pi}{3}$ (2)

iii. If $\alpha, \beta, \gamma$ are direction angles of a line and $\alpha = 60°$, $\beta = 45°$, then $\gamma$ = _____.

(A) 30° or 90° (B) 45° or 60° (C) 90° or 130° (D) 60° or 120° (2)

iv. The perpendicular distance of the plane $\bar{r} \cdot (3\hat{i} + 4\hat{j} + 12\hat{k}) = 78$, from the origin is _____.

(A) 4 (B) 5 (C) 6 (D) 8 (2)

v. The slope of the tangent to the curve $x = \sin\theta$ and $y = \cos 2\theta$ at $\theta = \frac{\pi}{6}$ is _____.

(A) $-2\sqrt{3}$ (B) $\frac{-2}{\sqrt{3}}$ (C) $-2$ (D) $-\frac{1}{2}$ (2)

vi. If $\int_{\frac{-\pi}{4}}^{\frac{\pi}{4}} x^3 . \sin^4 x \, dx = k$ then k = _____.

(A) 1 (B) 2 (C) 4 (D) 0 (2)

vii. The integrating factor of linear differential equation $x\frac{dy}{dx} + 2y = x^2 \log x$ is _____.

(A) $x$ (B) $\frac{1}{x}$ (C) $x^2$ (D) $\frac{1}{x^2}$ (2)

viii. If the mean and variance of a binomial distribution are 18 and 12 respectively, then the value of n is _____.

(A) 36 (B) 54 (C) 18 (D) 27 (2)

**Q.2. Answer the following questions:** **[4]**

i. Write the compound statement 'Nagpur is in Maharashtra and Chennai is in Tamilnadu' symbolically. (1)

ii. If the vectors $2\hat{i}-3\hat{j}+4\hat{k}$ and $p\hat{i}+6\hat{j}-8\hat{k}$ are collinear, then find the value of p. (1)

iii. Evaluate: $\int \frac{1}{x^2+25}\,dx$ (1)

iv. A particle is moving along X-axis. Its acceleration at time t is proportional to its velocity at that time. Find the differential equation of the motion of the particle. (1)

## SECTION – B

**Attempt any EIGHT of the following questions:** **[16]**

**Q.3.** Construct the truth table for the statement pattern:
$[(p \rightarrow q) \wedge q] \rightarrow p$ (2)

**Q.4.** Check whether the matrix $\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ is invertible or not. (2)

**Q.5.** In $\Delta ABC$, if a = 18, b = 24 and c = 30 then find the value of $\sin\left(\frac{A}{2}\right)$. (2)

**Q.6.** Find k, if the sum of the slopes of the lines represented by $x^2 + kxy - 3y^2 = 0$ is twice their product. (2)

**Q.7.** If $\bar{a}, \bar{b}, \bar{c}$ are the position vectors of the points A, B, C respectively and $5\bar{a}-3\bar{b}-2\bar{c}=\bar{0}$, then find the ratio in which the point C divides the line segment BA. (2)

**Q.8.** Find the vector equation of the line passing through the point having position vector $4\hat{i}-\hat{j}+2\hat{k}$ and parallel to the vector $-2\hat{i}-\hat{j}+\hat{k}$. (2)

**Q.9.** Find $\frac{dy}{dx}$, if $y = (\log x)^x$. (2)

**Q.10.** Evaluate: $\int \log x\,dx$. (2)

**Q.11.** Evaluate : $\int_0^{\frac{\pi}{2}} \cos^2 x\,dx$. (2)

**Q.12.** Find the area of the region bounded by the curve $y = x^2$ and the lines $x = 1$, $x = 2$ and $y = 0$. (2)

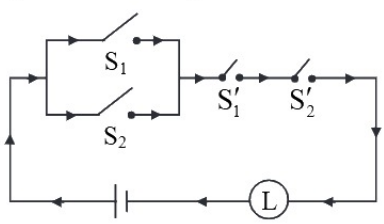
**Q.13.** Solve: $1+\frac{dy}{dx} = \text{cosec}(x+y)$ ; put $x + y = u$. (2)

**Q.14.** If two coins are tossed simultaneously, write the probability distribution of the number of heads. (2)

## SECTION – C

**Attempt any EIGHT of the following questions:** **[24]**

**Q.15.** Express the following switching circuit in the symbolic form of logic. Construct the switching table:




(3)

**Q.16.** Prove that: $\tan^{-1}\left(\frac{1}{2}\right)+\tan^{-1}\left(\frac{1}{3}\right)=\frac{\pi}{4}$ (3)

**Q.17.** In ΔABC, prove that: $\frac{\cos A}{a}+\frac{\cos B}{b}+\frac{\cos C}{c}=\frac{a^2+b^2+c^2}{2abc}$. (3)

**Q.18.** Prove by vector method, the angle subtended on a semicircle is a right angle. (3)

**Q.19.** Find the shortest distance between the lines $\bar{r}=\left(4\hat{i}-\hat{j}\right)+\lambda\left(\hat{i}+2\hat{j}-3\hat{k}\right)$ and $\bar{r}=\left(\hat{i}-\hat{j}-2\hat{k}\right)+\mu\left(\hat{i}+4\hat{j}-5\hat{k}\right)$. (3)

**Q.20.** Find the angle between the line $\bar{r}=\left(\hat{i}+2\hat{j}+\hat{k}\right)+\lambda\left(\hat{i}+\hat{j}+\hat{k}\right)$ and the plane $\bar{r}\cdot\left(2\hat{i}+\hat{j}+\hat{k}\right)=8$. (3)

**Q.21.** If $y = \sin^{-1}x$, then show that: $(1-x^2)\ \frac{d^2y}{dx^2}-x\cdot\frac{dy}{dx}=0$. (3)

**Q.22.** Find the approximate value of $\tan^{-1}(1.002)$.
[Given: $\pi = 3.1416$] (3)

**Q.23.** Prove that: $\int\frac{1}{a^2-x^2}\ dx=\frac{1}{2a}\log\left(\frac{a+x}{a-x}\right)+c$ (3)

**Q.24.** Solve the differential equation:
$x.\ \frac{dy}{dx}-y+x\cdot\sin\left(\frac{y}{x}\right)=0$ (3)

**Q.25.** Find k, if
$f(x) = kx^2(1-x)$, for $0 < x < 1$,
$= 0$ otherwise
is the p.d.f. of random variable X. (3)

**Q.26.** A die is thrown 6 times, if 'getting an odd number' is success, find the probability of 5 successes. (3)

## SECTION – D

**Attempt any FIVE of the following questions:** **[20]**

**Q.27.** Solve the following system of equations by the method of reduction:
$x + y + z = 6, y + 3z = 11, x + z = 2y$ (4)

**Q.28.** Prove that the acute angle θ between the lines represented by the equation $ax^2 + 2hxy + by^2 = 0$ is
$\tan\theta=\left|\frac{2\sqrt{h^2-ab}}{a+b}\right|$.
Hence find the condition that the lines are coincident. (4)

**Q.29.** Find the volume of the parallelopiped whose vertices are A(3, 2, –1), B(–2, 2, –3), C(3, 5, –2) and D(–2, 5, 4). (4)

**Q.30.** Solve the following L.P.P. by graphical method:
Maximize: $z = 10x + 25y$
Subject to : $0 \le x \le 3$,
$0 \le y \le 3$,
$x + y \le 5$.
Also find the maximum value of z. (4)

**Q.31.** If $x = f(t)$ and $y = g(t)$ are differentiable functions of t, so that $y$ is function of $x$ and $\frac{dx}{dt}\neq 0$ then prove that $\frac{dy}{dx}=\frac{\frac{dy}{dt}}{\frac{dx}{dt}}$.
Hence find $\frac{dy}{dx}$, if $x = at^2, y = 2at$. (4)

**Q.32.** A box with a square base is to have an open top. The surface area of box is 147 sq. cm. What should be its dimensions in order that the volume is largest? (4)

**Q.33.** Evaluate: $\int \frac{5e^x}{(e^x+1)(e^{2x}+9)}dx$ (4)

**Q.34.** Prove that: $\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx$

Hence show that: $\int_0^{\pi} \sin x \ dx = 2\int_0^{\frac{\pi}{2}} \sin x \ dx$ (4)